



- 6 A and B are two fixed points at a distance $22a$ apart, with B vertically below A . A light elastic string of natural length $4a$ and modulus of elasticity $5mg$ has one end attached to A and the other end attached to a particle P of mass km . Another light elastic string of natural length $8a$ and modulus of elasticity $6mg$ has one end attached to B and the other end attached to P . Particle P is vertically above B .

(a) Show that, when the system is in equilibrium, $BP = \frac{57a - 2ak}{4}$. [5]

[illegible]



The particle P is pulled vertically upwards so that $BP = 18a$, and is then released from rest. In its subsequent motion, P first comes to instantaneous rest at the point where $BP = 8a$.

- (b) Find the value of k . [4]

[illegible]